

■ CRIME INTELLIGENCE & ANALYTICAL PLATFORM

KSP Datathon 2026 — Hack2Skill

Track 2: AI-Driven Crime Analytics & Visualization Platform

Prize Pool	■10,00,000
Registration Deadline	July 19, 2026
Grand Finale	In-person, Bengaluru
Host	Karnataka State Police / SCRB

"Transform the Karnataka State Police's fragmented, Excel-based crime records into a real-time, AI-powered Strategic Intelligence Hub — enabling proactive policing, network analysis, and spatiotemporal crime visualization across all 31 districts."

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■ Crime Intelligence & Analytical Platform (Lumina)

KSP Datathon 2026 — Karnataka State Police × Hack2Skill

Track 2: AI-Driven Crime Analytics & Visualization Platform

Deployment: 100% Zoho Catalyst Native

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1. Project Overview

Field	Details
Platform Name	Lumina (Crime Intelligence & Analytical Platform)
Hackathon	KSP Datathon 2026 — Hack2Skill
Track	Track 2 — AI-Driven Crime Analytics & Visualization
Host	Karnataka State Police (KSP) / SCRB
Deployment Requirement	Zoho Catalyst (Mandatory)

Vision Statement

Transform the Karnataka State Police's fragmented, Excel-based crime records into a **real-time, AI-powered Strategic Intelligence Hub** deployed seamlessly on **Zoho Catalyst** — enabling proactive policing, network analysis, and spatiotemporal crime visualization.

2. Problem Analysis

[Architecture Diagram — MERMAID]

3. Solution Architecture (Catalyst)

3.1 System Architecture Diagram

[Architecture Diagram — MERMAID]

4. Tech Stack (Strict Catalyst Mapping)

Standard Tech	Replaced By	Purpose in Lumina
React / Nginx	Catalyst Web Client Hosting	Hosting the SPA frontend
FastAPI / Docker	Catalyst Serverless (Functions)	Core backend REST API logic
Keycloak	Catalyst Authentication	User login, signup, RBAC
Kong API Gateway	Catalyst API Gateway	API routing, throttling, protection
PostgreSQL	Catalyst Data Store	Relational DB for FIRs, Suspects
MinIO / S3	Catalyst Stratus	Object storage for raw CSVs and PDFs
Redis	Catalyst Cache	Fast query caching for dashboards
Apache Airflow	Catalyst Cron / Job Scheduling	Nightly batch processing

Standard Tech	Replaced By	Purpose in Lumina
Apache Kafka	Catalyst Signals	Event-driven architecture (file uploads)
OpenAI / LangChain	Catalyst QuickML (LLM/RAG)	AI chat interface for natural language queries
spaCy	Catalyst Zia Services	Text analytics for FIR entity extraction
React-PDF	Catalyst SmartBrowz	Server-side PDF report generation
WebSockets	Catalyst Push Notifications	Real-time anomaly alerts
Docker / K8s	Catalyst AppSail	Running Neo4j and custom ML pipelines

5. Data Architecture & Schema

5.1 Catalyst Data Store Schema (Relational)

Tables:

- 1. **District** (ID, Name, Code, Population)
- 2. **Police_Station** (ID, District_ID, Name, Jurisdiction_Area)
- 3. **FIR** (ID, Station_ID, FIR_Number, Date, Crime_Group, Lat, Lon, Narrative, Status)
- 4. **Accused** (ID, Name, DOB, Gender, Occupation, Arrest_Count)
- 5. **Victim** (ID, Name, DOB, Gender, Socioeconomic_Status)
- 6. **Case_Accused** (Mapping: FIR_ID, Accused_ID, Involvement_Type)
- 7. **Risk_Score** (ID, District_ID, Crime_Type, Score, Forecast_Date)

5.2 Neo4j Graph Schema (Hosted on AppSail)

Since network topology requires multi-hop recursive queries, Neo4j is deployed via Catalyst AppSail to run GraphSAGE link prediction.

(:Suspect)-[:COMMITTED]->(:Incident)
(:Victim)-[:VICTIMIZED_IN]->(:Incident)
(:Incident)-[:OCCURRED_AT]->(:Location)
(:Suspect)-[:ASSOCIATED_WITH]->(:Suspect)

6. AI/ML Models Specification

6.1 Catalyst Zia & QuickML Integration

Model	Catalyst Service	Input	Output
Predictive Risk Scorer	Catalyst Zia AutoML	Tabular historical density, time features	Risk score 0–100 per district
NER Extractor	Catalyst Zia Services (Text)	Raw FIR narrative text	Persons, locations, weapons
Query Assistant	Catalyst QuickML (RAG)	Natural language query	Structured response & SQL filter
Hotspot Detector	AppSail (Python ML)	(lat, lon, timestamp, crime_type)	ST-DBSCAN GeoJSON clusters

7. Feature Breakdown

Feature 1: State Overview Map

- **Technology:** Deck.gl `HeatmapLayer` on Catalyst Web Client Hosting.
- **Data:** FIRs fetched via Catalyst Serverless from Data Store.

Feature 2: Hotspot Explorer (ST-DBSCAN Clusters)

- **Technology:** Deck.gl `PolygonLayer` with pulsing animation.
- **AI:** ST-DBSCAN running on AppSail; clusters served via API Gateway.

Feature 3: Criminal Network Graph

- **Technology:** Cytoscape.js with Cose-Bilkent layout.
- **Data:** Neo4j queried via AppSail container.

Feature 4: Predictive Risk Score Board

- **Technology:** ECharts heatmap grid.
- **AI:** Catalyst Zia AutoML forecasting crime risk for the next 7/14 days.

Feature 5: AI Query Assistant

- **Technology:** Chat UI communicating with Catalyst QuickML.

- **Action:** Officers query "Show robbery trends in Mysuru" in natural language.

Feature 6: One-Click Briefing Reports

- **Technology:** Catalyst SmartBrowz.
 - **Action:** Instantly generates PDF reports of current dashboard state for offline field use.
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8. Development Roadmap

Sprint 1 — Catalyst Foundation (Days 1–2)

- Set up Zoho Catalyst Project via Catalyst CLI.
- Define schemas in **Catalyst Data Store**.
- Create synthetic data generator and upload to **Catalyst Stratus**.
- Configure **Catalyst Authentication** (RBAC roles).

Sprint 2 — Core APIs & AI (Days 3–4)

- Write **Catalyst Serverless** functions (Advanced I/O) for CRUD operations.
- Deploy Neo4j Docker image to **Catalyst AppSail**.
- Integrate **Catalyst Zia Services** for NER on FIR texts.
- Setup **Catalyst QuickML** for the RAG chat assistant.

Sprint 3 — Frontend (Days 5–6)

- Build React + Deck.gl application.
- Deploy to **Catalyst Web Client Hosting**.
- Integrate **Catalyst API Gateway** for secure frontend-backend communication.
- Implement Cytoscape.js network graphs.

Sprint 4 — Polish & Demo (Days 7)

- Setup **Catalyst SmartBrowz** for PDF generation.
 - Setup **Catalyst Cron** for scheduled ML updates.
 - Prepare demo script ensuring every Catalyst service is highlighted to judges.
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9. Deployment Strategy

The entire platform is deployed using the **Catalyst CLI** (`catalyst deploy`). No external cloud providers (AWS/GCP/Azure) are used, guaranteeing compliance.

project-root/
■■■ catalyst.json
■■■ client/ # Deployed to Web Client Hosting
■■■ functions/
■ ■■■ api_service/ # Advanced I/O Serverless Function
■ ■■■ etl_cron/ # Cron Function
■■■ appsail/
■■■ neo4j/ # Dockerfile for Graph DB
■■■ ml_pipeline/ # ST-DBSCAN Python runtime

10. Evaluation Alignment

Judging Criterion	Lumina Implementation	Catalyst Service Used
Platform Integration	100% Zoho Catalyst native deployment	All
Operational Usability	One-click PDF reports for field officers	SmartBrowz
AI/Analytics Quality	Tabular forecasting & LLM querying	Zia AutoML, QuickML
Data Handling	Schema-based synthetic data generation	Data Store, Stratus
Innovation	GNN criminal network + spatial clustering	AppSail